

Literature Review

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Article 1:

An evaluation of mapped species distribution models used for conservation planning.

Authors: Johnson and Gillingham (2005)

Background

- Researchers compare the performance of species distribution models (SDMs) using four different modeling techniques: Habitat suitability index (HSI), logistic regression-based resource selection function (RSF), Mahalanobis distance, and an ecological niche model.
 - HSI applies expert knowledge and meta-analysis to rank wildlife-habitat relationships using additive, multiplicative or logical variables with coefficients typically ranging from 0 to 1 based on perceived relevance.
 - RSF was defined as “any mathematical function that provides predictions of resource or habitat use that are proportional to the true probability of use.” The model is based on the theory that when use of habitat exceeds availability, then selection is inferred, and when use is less than availability then the habitat is avoided.
 - Mahalanobis distance is described as a multivariate dissimilarity statistic. In the context of SDMs, the mean conditions of a set of environmental variables are compared to other locations around the broader landscape. When a larger value of the statistic represents a larger deviation from the mean conditions.
 - Ecological niche models are explicitly linked to niche theory, and are used to represent species distribution across larger scale regions. These models seem to require much more in-depth knowledge about the studied species as the model relies

on evolutionary constructs that determine whether a set of environmental variables can or cannot support a species population.

- To evaluate performance, they measured differences in the distribution of ranked habitats, and they evaluated the predictive ability of the occurrence of habitats that correlated with known species locations.

Methods

- The authors selected models based on relevance (i.e. contemporary and broad use in existing literature), ability to be used within GIS, and representation of the data types necessary to parameterize models.
 - The HSI was based on previous research and was derived from published literature, agency reports and expert opinion. The index was then ranked on a six class scale: high, moderately high, moderate, low, very low, and nil.
 - The RSF was based on fixed-effects logistic regression:

$$w(x) = \exp(\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)$$

where $\beta_1 \dots \beta_n$ are coefficients generated from the regression and $w(x)$ is the relative probability of species occurrence.

- Mahalanobis distance was calculated at each raster cell using GIS using the formula:

$$D^2 = (x - m)^T C^{-1} (x - m)$$

where D^2 is the Mahalanobis distance, x is the habitat variable, m is the mean value of habitat characteristics recorded at known locations, T indicates that vector should be transposed, and C^{-1} is the covariance matrix for the same vector of habitat characteristics. The authors used ESRI ArcView and GIS IDRISI to perform matrix calculations on ecological variables.

- The ecological niche model was adopted from the Genetic Algorithm for Rule-set Prediction (GARP) computational approach for modeling. The GARP model is a standalone machine-learning process that develops rule sets to best identify species distribution.
- For the RSF, Mahalanobis distance, and niche model, the authors developed 3 sets of models for each. The HSI model was predetermined via previous research. A total of 10 maps were created and normalized by calculating number of species locations within each of 6 habitat classes.

- Relationship between predicted class and normalized frequency of species locations were evaluated using Spearman rank and Pearson correlations. Kappa index of agreement was used to measure spatial correlation between maps.
- Using the results of the correlations, the authors evaluated only the maps for each model that demonstrated the highest predictive ability.

Key Findings

- There was no reported significance in predictive capability between different variable sets or quantitative models. Comparing quantitative to qualitative models, the maps generated with HSI had weakest correlation and underperformed relative to other techniques.
- Of the models, Mahalanobis distance was best at predicting high-quality habitat, followed by RSF and niche models. Conversely, HSI was poor at predicting high-quality habitat. All models performed well at predicting low-quality habitat.
- Modeling technique largely determined by data availability and costs. RSFs provide greatest flexibility as model parameters are easily assessed. Mahalanobis distances and niche models do not provide metrics for evaluating strength of variable relationships. HSI coefficients are difficult to evaluate.

Significance

- Author's demonstrate that there is a benefit to using quantitative models over qualitative models when evaluating species distribution. But acknowledge, that the choice of modeling technique should be based on data availability and costs. Further, this paper demonstrates that there is value in using multiple modeling techniques to compare the effectiveness for SDM.

Relevance to Capstone and Final Thoughts

- This paper largely employs the same thesis that my project would be based around. I think there could be some potential using the same approach to evaluate Oyster habitat in the Pensacola Bay area.
- While this paper is very helpful in guiding my thoughts on my own research, I Will keep searching for more recent papers that have used similar approaches to evaluate species distribution as this one is from 2005.
- The author's cite a number of research papers that aim to compare model performance in SDM that would prove beneficial to review.

References

Johnson, Chris J., and Michael P. Gillingham. 2005. "An Evaluation of Mapped Species Distribution Models Used for Conservation Planning." *Environmental Conservation* 32 (June): 117–28. <https://doi.org/10.1017/S0376892905002171>.